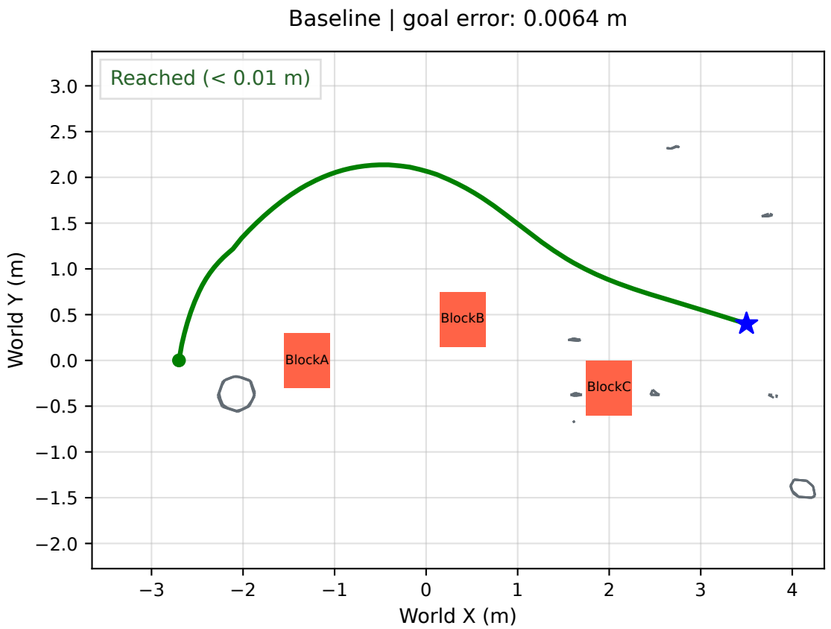
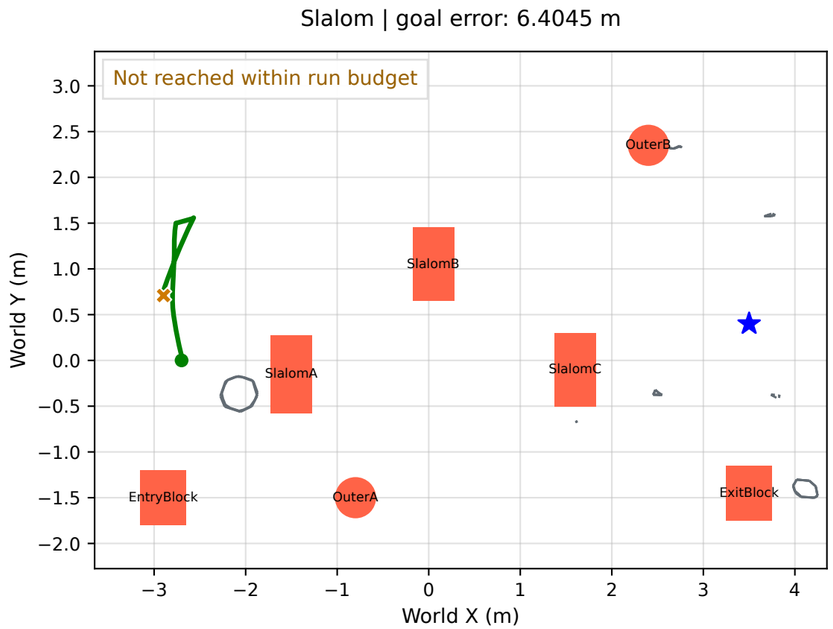
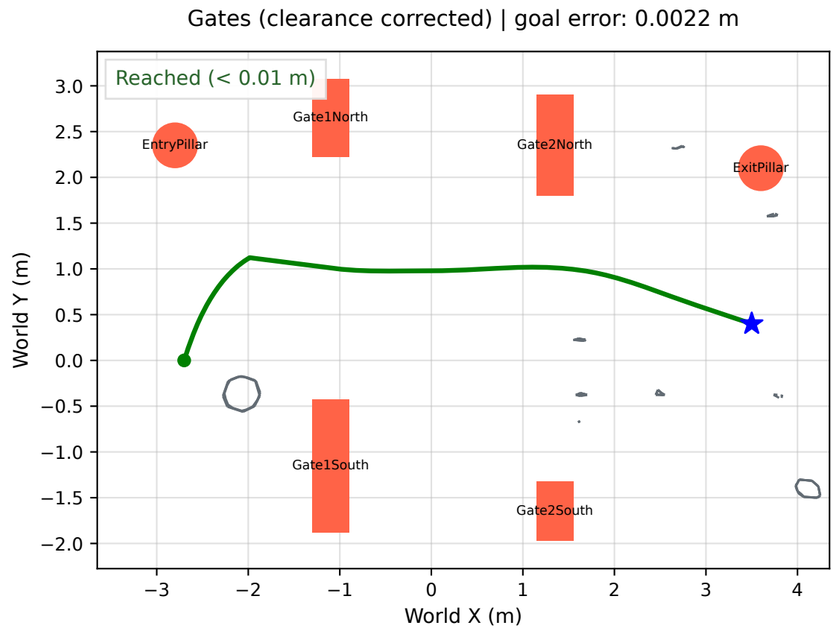
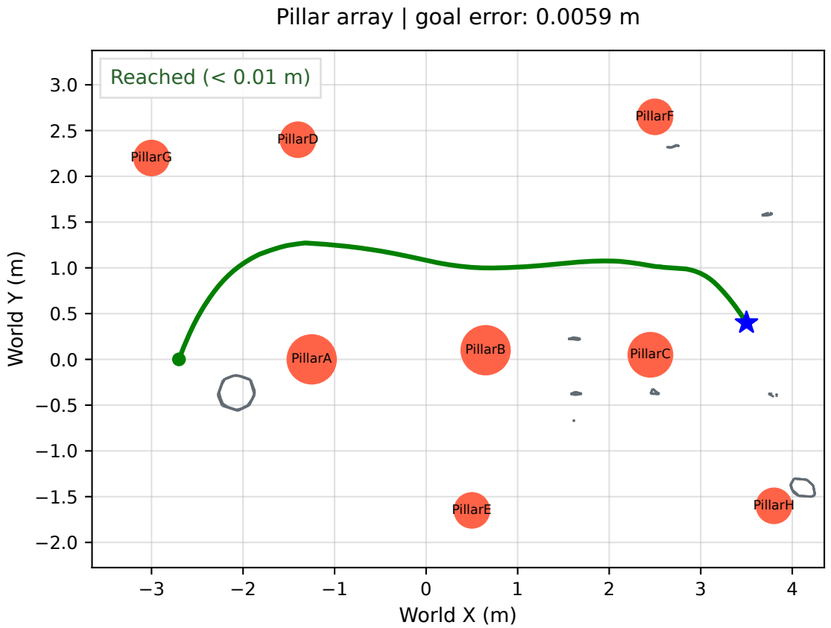
ROV obstacle-avoidance routes across five pool scenarios









Executed ROV path
Start
Goal
Added obstacle footprint
Original pool structures
Final position: not reached

Measured trajectories from one test suite

Goal: [3.5, 0.4, 1.0] m
Fixed reference height: z = 1.0 m
Ideal depth + ground-truth odometry
Arrival tolerance: 0.01 m; dwell: 2 s
Simulation budget: 180 s per scenario
Gates: corrected opening clearance

Routes are plotted without smoothing
or extending them to the goal.